INVESTMENT DECISION BRIEF | ILLUSTRATIVE CASE

Predictive maintenance for a mid-size factory

Mid-size factory, manufacturing | Prepared by Chaitali Shekar | October 2026

Recommendation: Proceed after fixes

Close the readiness gaps first ("fits how people work" and "people trust it"). It saves money even in the worst case ($164k a year).

PROJECTED SAVINGS A YEAR	PAYS FOR ITSELF IN	3-YEAR ROI	IF COSTS ARE WORSE
$660k	**4.7 months**	**1,147%**	**$164k**
after AI running costs	$150k to set up	$1.72M net over 3 years	savings a year, worst case

THE SITUATION

A mid-size parts maker has about 40 unplanned machine breakdowns a year at about $25k each in downtime and emergency repair: **$1.00M a year**. The machines already record temperature, speed, torque and tool wear, so a model can warn before a breakdown and turn it into a planned repair at about $4k. Predictive maintenance was the best fit of 12 AI options evaluated, and the only one aimed at unplanned downtime.

OPTIONS CONSIDERED

OPTION	BREAKDOWNS CAUGHT	FALSE ALARMS A YEAR	NET SAVINGS A YEAR	ASSESSMENT
Do nothing	0%	0	$0	Keeps losing $1.00M a year
Alerts from the maintenance manual's limits	98%	85	$721k	Only 31% of alerts are real. Strong here only because the synthetic data was generated from these same limits
AI warning model (selected)	86%	9	$660k	80% of alerts are real, which protects crew trust; the manual's limits are kept to explain each warning

HOW THE SAVINGS ARE WORKED OUT

EACH YEAR	AMOUNT	BASIS
Breakdowns today	$1.00M	40 breakdowns x $25k
Warned in time, repaired as planned	$138k	34.5 x $4k planned repair
Missed breakdowns	$137k	5.5 x $25k
False alarms	$4k	8.6 inspections x $500
Running the AI	$60k	licence, computing and support
Net savings	**$660k**	**breakdowns today minus the four costs above**

RANGE OF OUTCOMES

CASE	NET SAVINGS A YEAR	3-YEAR NET VALUE	PAYBACK
Costs worse than expected	$164k	$212k	20.5 months
As planned	$660k	$1.72M	4.7 months
Costs better than expected	$1.33M	$3.77M	2.0 months

Biggest driver: cost of an unplanned failure, which moves 3-year value by $1.96M. Worse case = every cost assumption at the end of its range that hurts value.

READINESS GAPS AND RISKS

RISK	EFFECT	MITIGATION	OWNER
Fits how people work (2.0 / 5)	Warnings do not reach the crew; savings fall short	Map how a warning becomes a repair, and send warnings straight into the maintenance system before launch.	Maintenance manager
People trust it (2.5 / 5)	The crew ignores correct warnings	Design the warnings with technicians, show why each one appears, and start with one production line.	Operations director
Cost of a breakdown is an estimate	Moves 3-year value by $1.96M	Agree it with finance from repair invoices before approval	Finance
The dataset is synthetic	Accuracy in a real plant is likely lower; simple rules look unrealistically good	Pilot on one line and measure catch rate and false alarms before scaling	Data / IT lead
The AI gets weaker over time	Fewer breakdowns caught	Monthly check; update the AI if its catch rate drops 10+ points over 3 months	Data / IT lead

DECISION RULE

RECOMMENDATION	WHEN
Stop	Expected savings are not positive
Revise	Savings turn negative if costs are worse than expected, or the team is not ready
Proceed after fixes	Savings stay positive in the worst case, but the team check says fix first
Proceed	Savings stay positive in the worst case and the team is ready

PROPOSED PLAN AND DECISION GATES

WHEN	WHAT HAPPENS	GATE TO PASS
Month 0	Approve the $150k setup budget; agree the cost of a breakdown with finance	Savings stay positive in the worst case
Months 1-2	Connect warnings to the maintenance system, close the readiness gaps, train the crew	Readiness verdict is Go; all 13 acceptance tests pass
Launch	Warnings go live on one line; a planner approves every work order	No work order released without approval
Every month	Compare savings with the forecast and explain any gap	Crew acts on 70%+ of warnings; catch rate within 10 points of launch
Month 6	Decide whether to scale to more lines, improve, or reconsider	At least 90% of the forecast delivered to scale

EVIDENCE BEHIND THE NUMBERS

Measured: catch, miss and false-alarm rates (3,000 held-out machine records (AI4I 2020, synthetic)) | **Assumption:** plant costs (Illustrative defaults, editable in the app) | **Estimate:** ai option scores (My scoring of 12 options on 7 criteria) | **Assessment:** readiness scores (Example ratings for the factory) | **Simulated:** results after launch (12 made-up months run through the S3 to Databricks pipeline)